Thesis Progress Report – Year[1]

Name and first name of the doctoral candidate: MA JUN

Research unit: LMAC(Laboratoire de Mathématiques Appliquées de Compiègne)

Thesis supervisor(s): Faker Ben Belgacem & Faten Jelassi

This report helps you to present your thesis at the annual meeting of your Individual Monitoring Committee (CSI), both HR (Human Resources) and Scientific. You must forward this report to your CSI members at least 10 days prior to your CSI meeting date.
You must also upload this report in ADUM in your administrative re-registration file.

Here are at least the topics to fill in, you can add some if you wish:

Brief description of thesis topic and objectives:

The project is to develop innovative Thermal Energy Storage (TES) Hybrid systems by integrating topology optimization, shape optimization, reduced-order modeling, complemented by the potential resources (and applications) of Physical Informed Deep Learning Techniques. Specifically, the target is to enhance the thermal conductivity of TES systems that incorporate Phase Change Materials (PCMs), overcoming existing limitations through advanced design and numerical simulation strategies. The focus will be on encapsulating the TES material by thermal conductive enhancers to achieve more efficient heat transfer, so to speed up the filling in and (especially) the extracting of the heat into and from the storage media during the short cycles loading/storage/unloading. The common feeling is that both (loading and unloading) optimization problems may not (are not expected to?) result in similar or even close distributions of the capsules. If so, this issue has to be discussed to come up with an in-between composite media that enables an efficient heat supply and an easy heat removal. A possible way for a well informed decision for the optimum is to import tools from multi-objective optimization such as the Pareto principle, for example. Implementing topology optimization procedures: the candidate is expected to explore reduced-order models for an efficient capsule arrangements that maintain the high heat capacity of PCMs while enhancing overall conductivity. This will facilitate rapid numerical simulation and analysis of these complex systems, and PINN algorithms will therefore be investigated for their potential to predict system performance and optimize design configurations.

Summary of work done for the past year and background:

- Starting from the theoretical analysis of steady-state heat conduction, ignoring the time derivative term, the control equation was analytically derived and numerically verified, thereby clarifying the spatial distribution characteristics of the temperature field. On this basis, the research is extended to the problem of transient heat conduction. To address the increased complexity of transient terms, a systematic approach combining Laplace transform, hyperbolic function analysis and series expansion was adopted. The challenges of parameter sensitivity and heat flux evolution modeling under spatio-temporal coupling conditions have been mainly addressed. The numerical implementation was carried out using the open-source software FreeFEM++ (completed in both one-dimensional and two-dimensional forms).
- To further consider the thermal radiation effect, the radiative heat transfer term is incorporated into the control equation according to the Stefan-Boltzmann law, and a nonlinear system of partial differential equations is obtained. The mathematical analysis of the steady-state heat conduction problem in the double-layer domain was mainly studied. In this problem, there exists a nonlinear Kapitza contact resistance at the interface, thereby slowing down the heat transfer. A simple one-

[1] Indicate the thesis year at the end of which this report is written (e.g.: Year 1 => Progress report of 1st year of thesis).

dimensional example was used to illustrate the non-uniqueness of the solution. The Picard iterative method is adopted to calculate the solution as a fixed point of a specific algebraic mapping. Co-dimensional one-bifurcation analysis was conducted on these mappings, with the ratio of the thermal conductivity of the two layers of materials as the control parameter. Two types of (power-law) Kapitza conductivity can cause different types of bifurcation phenomena, including transcritical bifurcation, Pitshfork bifurcation, flip bifurcation and saddle node bifurcation.

- The relevant theory of topological derivatives is studied, and the theoretical derivations were carried out using the Dirichlet-To-Neumann operator. The inversion of the relevant FreeFEM++ program was achieved through the topological optimization problem of two-dimensional transient heat conduction.

Perspectives and projects on thesis work for the coming year:

- First, we will build a comprehensive one-dimensional bifurcation atlas for the scalar model: begin with a codimension-one continuation in the control parameter δ and systematically classify equilibria and their local bifurcations (saddle-node, flip, pitchfork) as well as period-k cascades; then explore non-unique unfoldings by varying additional parameters—such as the exponent q, the boundary data at the two ends, and the coefficients λ and μ — independently; finally, extend to higher-codimension settings where several parameters vary simultaneously and document the resulting multi-parameter structure.
- Second we will develop nonlinear elliptic/parabolic models with volumetric sources and state-dependent coefficients in the domain, together with interface/boundary laws; in three dimensions we have established an a-priori L-infinity bound via De Giorgi–Stampacchia iterations and will next treat the transient problem from a dynamical-systems viewpoint, proving existence (e.g., via Leray–Schauder), investigating uniqueness, characterizing global attractors and basins of attraction, and formulating feedback/optimal control strategies.
- Finally, a central focus of the upcoming work lies in the construction of a reduced-order modeling strategy based on Physics-Informed Neural Networks (PINNs). Unlike classical ROMs relying purely on projection (e.g., POD), the PINN approach enables embedding physical laws directly into the learning process. This direction aims to capture complex thermal behavior with minimal labeled data, offering a scalable and generalizable modeling framework for heat transfer problems involving capsules or nonlinear contact conditions. It is anticipated that this integration will significantly improve the balance between accuracy, interpretability, and computational efficiency.

Doctoral training:

[1] CP36 - Practical tools to give meaning to your work and solve problems (6h)
[2] CST02 : Méthodes pour la modélisation aléatoire (20h)
[3] LA91, niveau A1/A2, - Semestre de printemps 2025 (15h)
[4] CST02 Python scientifique et Jupyterlab (26h)
[5] CIMPA research school: Mathematics, Artificial Intelligence and Application 2025(48h)

Review of dissemination actions and scientific and technical productions: (specify: submitted or accepted)

* **Internal scientific presentations** (specify dates and audience), oral or poster

NO

* **Publications in journals with or without a reading committee** (specify title, authors, references)

ACL: Articles in international or national peer-reviewed journals listed in international databases (ISI Web of Knowledge, Scopus, Pub Med...).
ALC: Articles in refereed journals not listed in international databases.
ASCL: Articles in non-peer-reviewed journals.

NO

* **Communications to symposia/congresses with acts** (specify title, dates, place, title, authors and references)
ACTI: Communications with acts at an international congress.
ACTN: Communications with deeds at a national congress.
COM: Oral communication without acts at an international or national congress.
POST: Posted communications at an international or national conference.

* **Patent filings** **NO**

* **Awards and honors** **NO**

* **Participation in the organization of national/international symposia** **NO**

* **Involvement in the life of the institution, in bodies ...** **NO**

* **Public communication actions** (specify the name, date and place of the meeting and your type of action) (e.g. Science Day)

[1] **FreeFem Days: An Academic and Industrial Research Hybrid Conference (In-person or via Zoom)**, Paris, December 2024. *(Academic conference, Sorbonne University)*

[2] **Meeting of the ANR Project NumOpTES**, January 28, 2025. *(Academic conference, University of Bordeaux)*

[3] **Winter School: "Control, Optimization, and Artificial Intelligence for Machine Learning"**, 2025. *(Winter school, Tunisia)*

[4] **Fifth Session of the LMAC Monthly Doctoral and Postdoctoral Seminar**, *(Academic seminar with presentation, Université de Technologie de Compiègne)*